

Computer Networks — Wireshark

1. What is Wireshark?

Wireshark is a **network protocol analyzer** used to:

- Capture packets/frames going into and out of a computer.
- Observe network protocols in action.
- Analyze captured network traffic.

GATE Point: Wireshark captures **Link-Layer frames**.

2. Layering & Encapsulation

When an application sends an HTTP message, data moves through different layers.

Application Layer

↓

HTTP Message

↓

Transport Layer

↓

TCP Segment

↓

Network Layer

↓

IP Datagram

↓

Link Layer

↓

Link-Layer Frame

Important

HTTP Message → TCP Segment → IP Datagram → Link-Layer Frame

Each layer adds its own **header** to the data received from the upper layer.

3. How Wireshark Captures Packets

Wireshark works by obtaining a copy of frames from the **Link Layer / network interface**.

Network Interface

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Captured Frames

↓

PCAP File

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Wireshark Analyzer

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Protocol Analysis

PCAP

PCAP (Packet Capture) is used to store captured network packets/frames.

4. Important Wireshark Components

Component	Purpose
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Interface List	Shows available network interfaces
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Start Capture	Starts packet capturing
Stop Capture	Stops packet capturing
Packet List	Displays captured packets
Packet Details	Shows details of selected packet
Packet Bytes	Shows packet data in Hex/ASCII
Display Filter	Filters packets for analysis

5. Example: HTTP Packet

When a browser communicates with a web server:

Client → HTTP GET Request → Server

Client ← HTTP Response ← Server

A captured frame may contain:

Ethernet Frame

→ **IPv4 Datagram**

→ **TCP Segment**

→ **HTTP Message**

Wireshark allows us to expand each layer and inspect its fields.

6. Why HTTP May Not Be Visible?

If HTTP packets are not visible in the capture, possible reasons are:

- **VPN** is being used.
- Website uses **HTTPS** instead of HTTP.
- Browser/OS privacy settings are encrypting traffic.
- Browser is using **HTTP/3 (QUIC)**.

Remember ★

HTTP → Application Layer

HTTPS → Encrypted HTTP traffic

HTTP/3 → Uses QUIC + encrypted traffic

★ GATE Quick Revision

Wireshark → Network protocol analyzer

Capture → Link-layer frames

PCAP → Packet capture file

HTTP → Application layer

TCP → Transport layer

IP → Network layer

Ethernet → Link layer

Display Filter → Filters displayed packets

💧 Most Important GATE Line

Wireshark captures Link-Layer frames, which contain encapsulated higher-layer protocols such as IP, TCP and HTTP.

Computer Networks — Cookies & Web Tracking

1. HTTP is Stateless

HTTP is a stateless protocol.

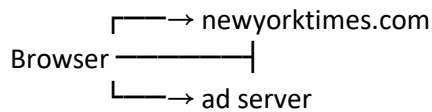
- Server normally does not remember previous requests automatically.
- **Cookies** are used to maintain/remember information about a user's interactions across visits.

- They can improve user experience, e.g. remembering login status or language preference.
GATE Point ★: Cookies provide a mechanism to maintain **state** over HTTP.

2. How a Web Page Can Contact Multiple Servers

A webpage may contain content from different servers.

For example:



Process

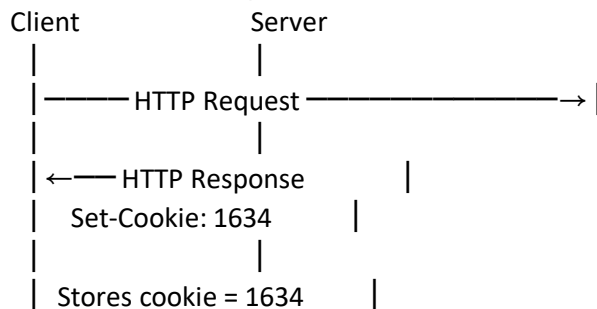
1. Browser requests the main webpage.
2. Main server sends the **HTML file**.
3. Browser parses the HTML.
4. If HTML contains a reference to another server, browser sends another HTTP request to that server.
5. The returned content is combined with the original page.

So, **one webpage can cause HTTP requests to multiple websites/servers**.

3. Cookies

A **cookie** is a small piece of information stored by the browser and associated with a particular website/domain.

Basic Working



The server can create a cookie value and store associated information in its database.

The server sends the cookie to the browser using the **Set-Cookie** header.

The browser stores the cookie and associates it with that website.

4. First-Party Cookie

Definition

A **first-party cookie** is issued by the website that the user **explicitly chose to visit**.

Example

User



newyorktimes.com



Cookie created by newyorktimes.com

The cookie is associated with the first-party website.

Use: Remembering login, preferences, previous interactions, etc.

Remember:

First party = Website you intentionally visited

5. Third-Party Cookie ★ ★ ★

A **third-party cookie** is issued by a website that the user **did not explicitly choose to visit**.

Example

A webpage contains an advertisement from another domain:

User



Website A



— Embedded Ad —→ Website B



Third-party cookie

When the browser requests the advertisement from Website B, Website B can send a cookie.

The browser stores this cookie even though the user did not directly visit Website B.

GATE Difference

First-Party Cookie

From website user explicitly visits

Used for site-specific state

Example: login/preferences

Third-Party Cookie

From another/embedded website

Can enable cross-site tracking

Example: advertising/tracking

6. Cookie-Based Tracking

This is the **most important concept** from this topic.

Suppose an advertising company gives the browser a unique cookie:

Cookie ID = 7493

Website 1

User visits:

news website → Ad server

Ad server sees cookie 7493.

It records:

Cookie 7493 → visited News Website

Website 2

Later user visits another website containing the same ad server.

Browser sends:

Cookie = 7493

The ad server recognizes:

"I have seen this browser before."

It can now associate multiple website visits with the same cookie value.

Cookie 7493



— News website

— Shopping website

— Other websites

Thus, third-party cookies can be used to **track browsing behavior across multiple websites**.

7. Targeted Advertising

Once an advertising server collects browsing information, it can use that information to provide **targeted advertisements**.

Example

User visits:

News site → Sports page



Shopping site → Socks



Ad server learns interests



Targeted advertisement

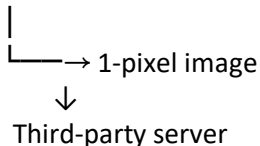
So:

Third-party cookie → User/browser identification → Browsing history → Targeted Ads

8. Third-Party Website Can Be Hidden

A browser may contact a third-party server even when the user does not visibly see that website. For example, a webpage can contain a very small **1-pixel image** whose URL belongs to another website.

Main Website



The browser fetches the image, causing a request to the third-party server.

Therefore, the user may **not even realize** that a third-party website was contacted.

9. Privacy Concern

Third-party cookies can create privacy concerns because they may allow companies to:

- Track browsing behavior across websites.
- Build a profile of browsing interests.
- Deliver targeted advertisements.

Important distinction

Initially, a cookie may identify a **browser**, not necessarily a person.

But if it is combined with other information such as:

- Login credentials
- IP address
- Other identifying information

it may be possible to associate the cookie with a **specific individual**.

10. GDPR

GDPR = General Data Protection Regulation

The transcript discusses the **European Union's GDPR**, introduced in **2018**.

GDPR treats information that can be used to identify an individual as **personal data**.

Cookies can fall under personal-data regulations when they can be associated with an individual.

Websites may therefore ask users for **explicit consent** regarding cookie usage.



GATE Quick Revision

Must Remember

HTTP → Stateless

Cookies → Maintain/remember state

Set-Cookie → Server sends cookie to browser

First-party cookie → From the website user explicitly visits

Third-party cookie → From another website embedded/referred by the page

Third-party cookies → Can enable cross-site tracking

Cookie + browsing data → Targeted advertising

1-pixel image → Can secretly trigger a third-party request

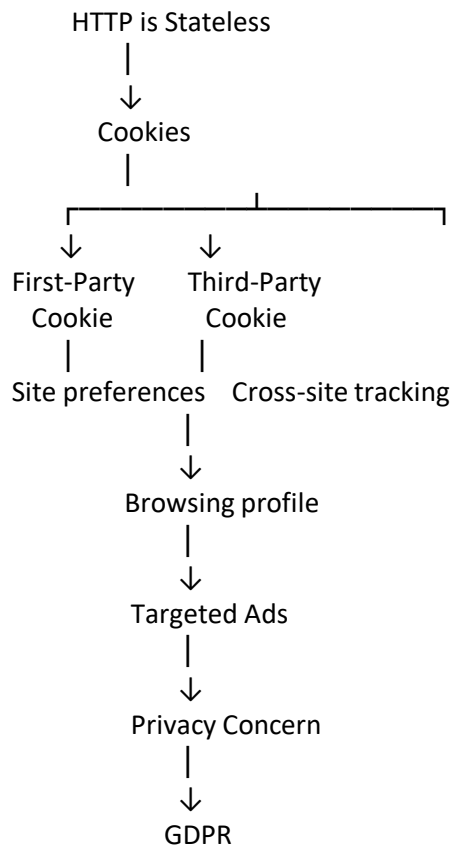
Cookie initially → May identify browser

Cookie + other identifying information → May identify individual

GDPR → Privacy/personal-data regulation



GATE Concept Map



Exam line:

Third-party cookies can be used to track a browser's activity across multiple websites and support targeted advertising.

~Anshu